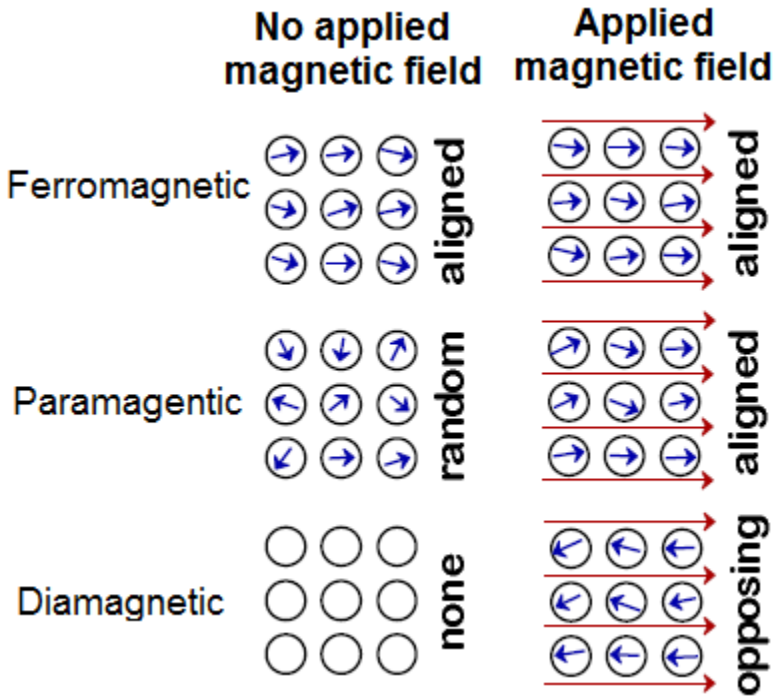
Physical Chemistry of Materials

Lecture 15

Applications of Solid-State Chemistry

Last Class

Magnetism



(Image: https://www.researchgate.net/publication/254705371_Understanding_the_Interaction_Between_Blood_Flow_and_an_Applied_Magnetic_Field)

Last Class

Independent Spin Model

Consider a solid system consisting of N spins:



$$E(\uparrow) = +\mu_{\mathrm{B}} B, \quad E(\downarrow) = -\mu_{\mathrm{B}} B$$

The system consists of distinguishable particles:

$$Z = z^N = [2\cosh(\beta \mu_{\mathrm{B}} B)]^N$$

(no phase transition)

Last Class

Ising Model

Consideration of neighboring spin-spin interactions

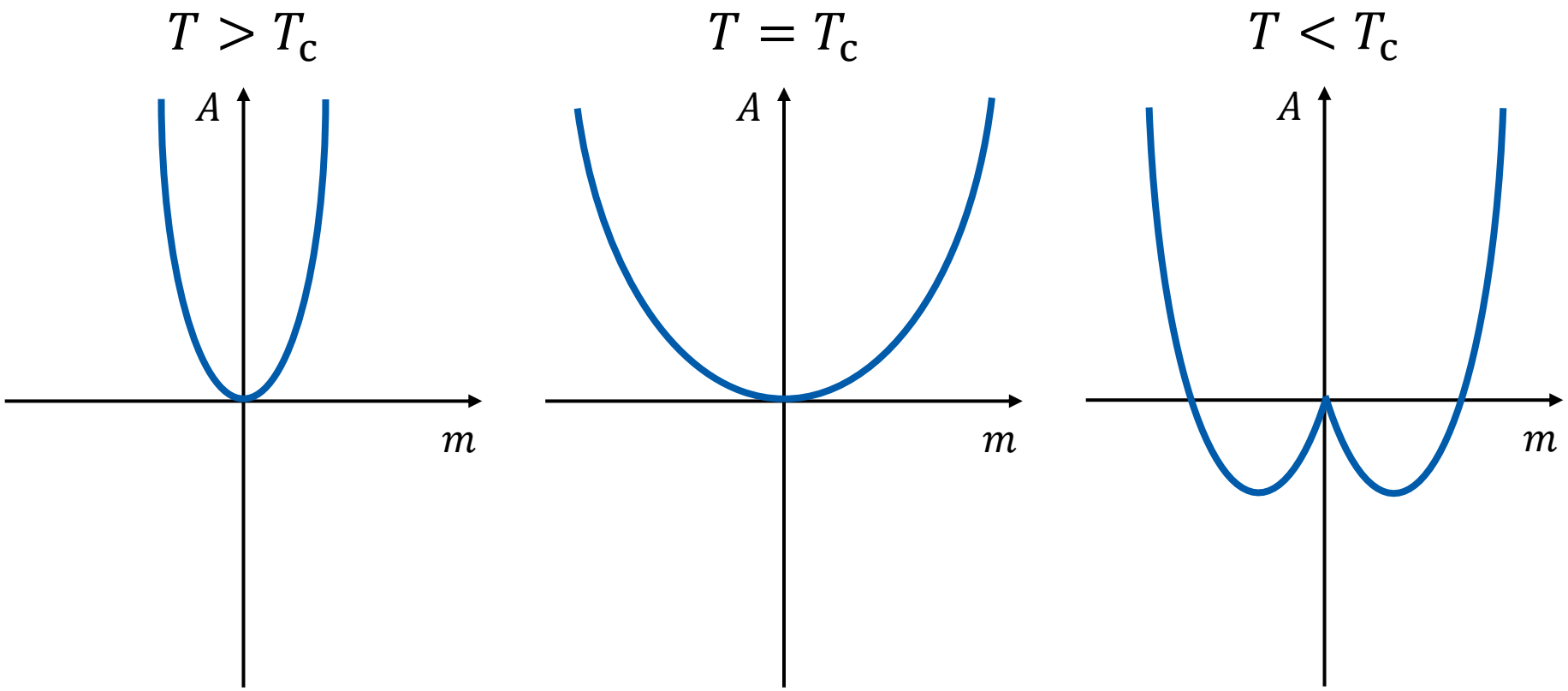
- System energy

$$E = \mu_{\mathrm{B}} B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

- Sign of J
  - $J > 0$: ferromagnetic
  - $J < 0$: antiferromagnetic

Last Class

Phase Transition in MF Model



1-Dimensional Ising Model

For $B = 0$,

$$Z = \sum_{\{\sigma_i\}} e^{\beta J \sum_{\langle ij \rangle} \sigma_i \sigma_j}$$

Consider the periodic boundary condition ($\sigma_{N+1} = \sigma_1$);

$$Z = \sum_{\sigma_1 = \pm 1} \sum_{\sigma_2 = \pm 1} \sum_{\sigma_3 = \pm 1} \cdots \sum_{\sigma_N = \pm 1} e^{\beta J \sum_i \sigma_i \sigma_{i+1}}$$

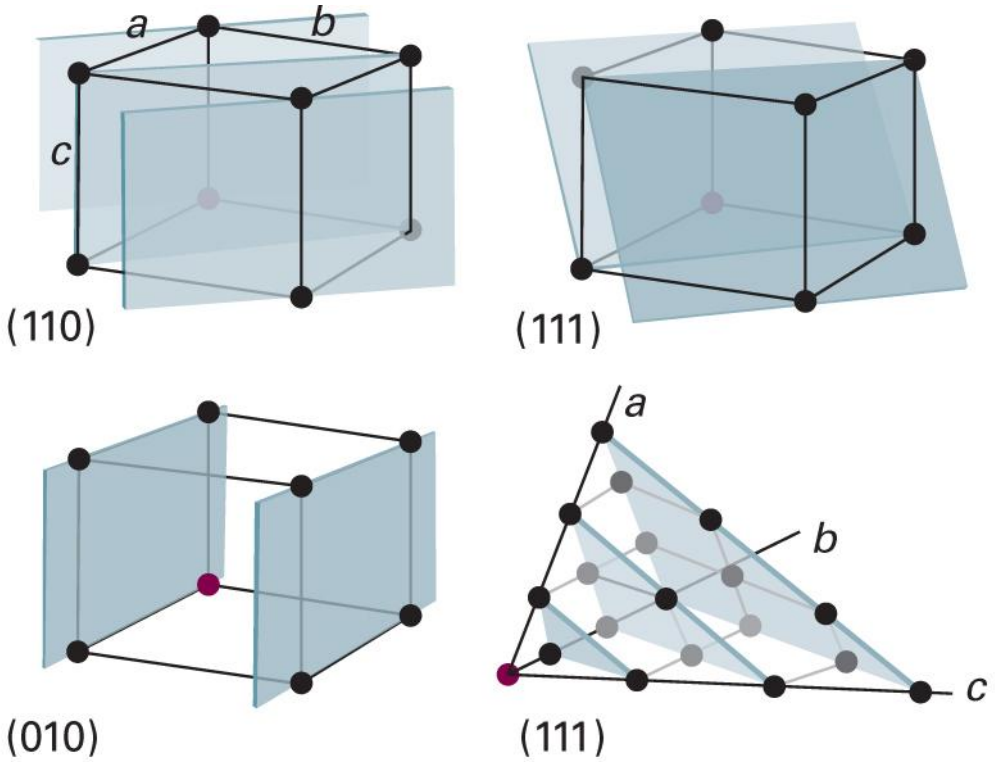
$$Z = [2\cosh(\beta J)]^N$$

(no phase transition)

Today's Topics

1. Surfaces: pp. 8-16
2. Defects and doping: pp. 17-21
3. Band bending: pp. 22-30
4. Nanomaterials: pp. 31-41

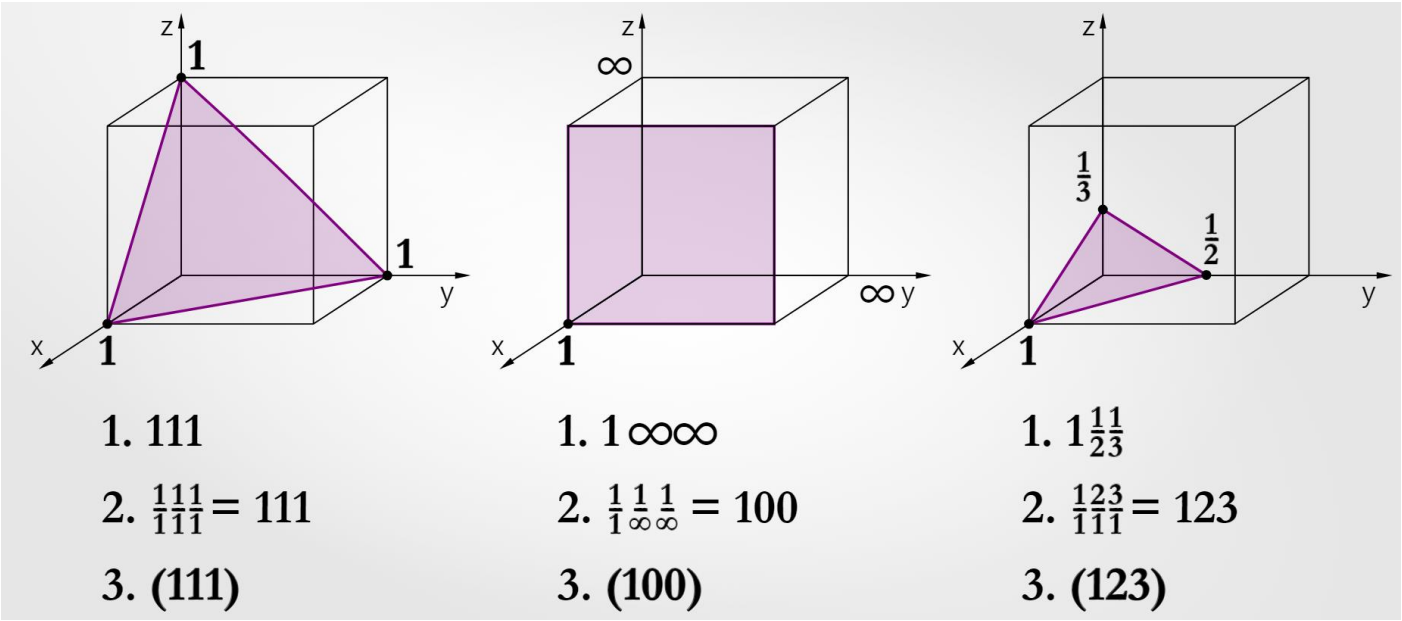
Lattice Planes



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 15A.12)

Miller Indices

Reciprocals of the intersection distances along lattice axes

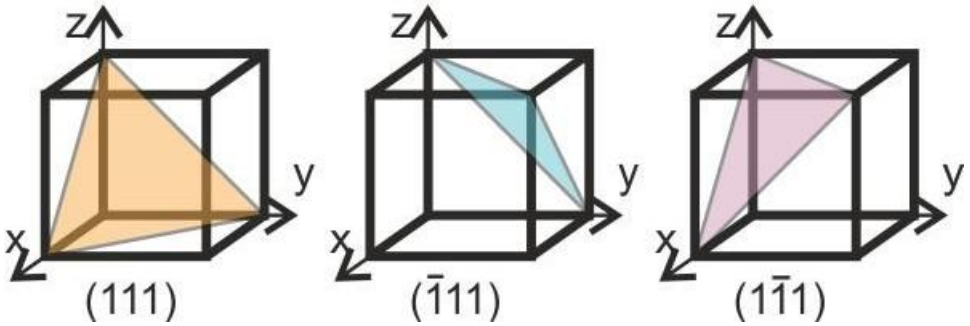


Integers!

(Image: https://msestudent.com/miller-indices/)

Notations

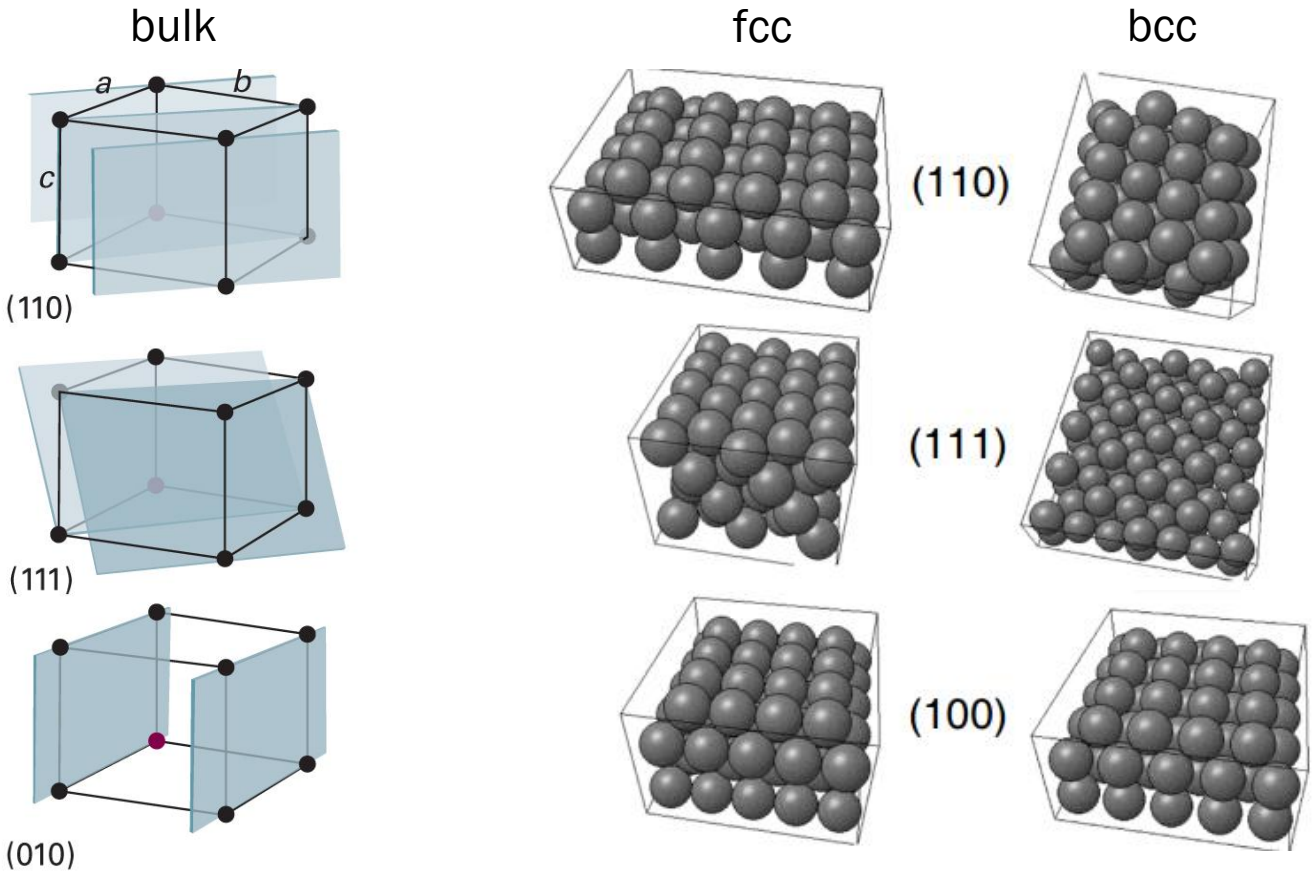
- Negative index: a bar over the number



- Brackets
  - (hkl): individual plane
  - $\{hkl\}$: set of parallel planes

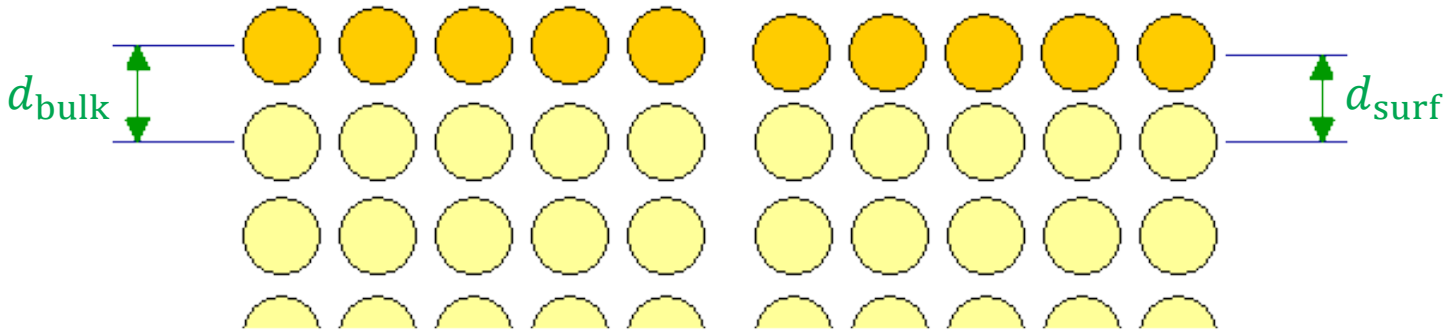
(Image: https://shorturl.at/TZdZV)

Ideal Crystal Surfaces



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 15A.11
https://www.chemie-biologie.uni-siegen.de/makro/master_surf_chem/lecture_introsurfchem_1b.pdf)

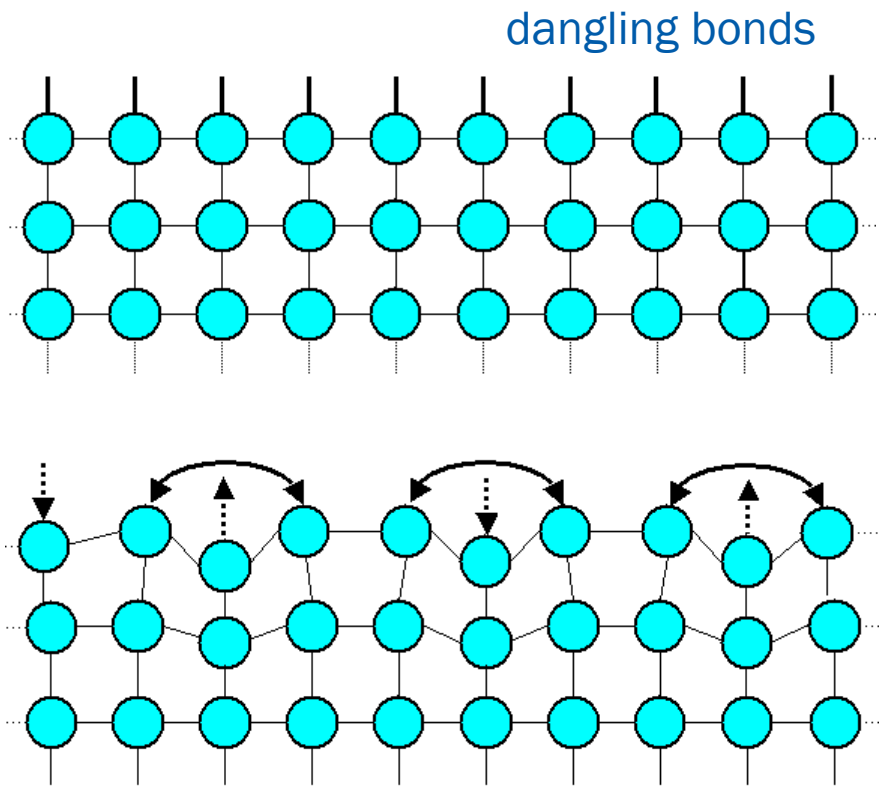
Surface Relaxation



$$d_{\text{bulk}} > d_{\text{surf}}$$

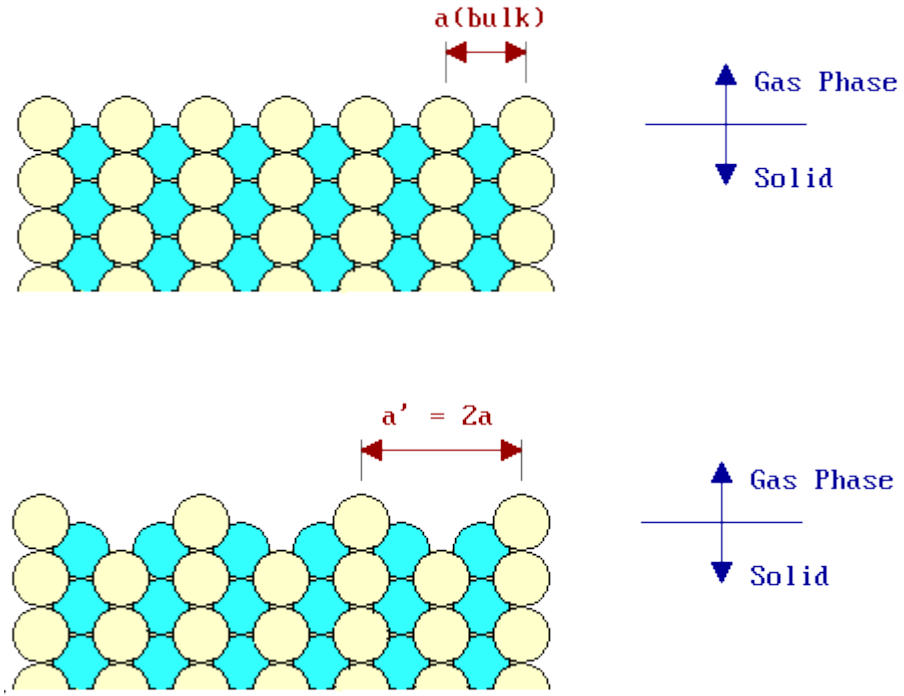
(Image: https://chem.libretexts.org/@go/page/25360)

Surface Reconstruction



(Image: https://www.tf.uni-kiel.de/matwis/amat/semitech_en/kap_3/advanced/t3_3_1.html)

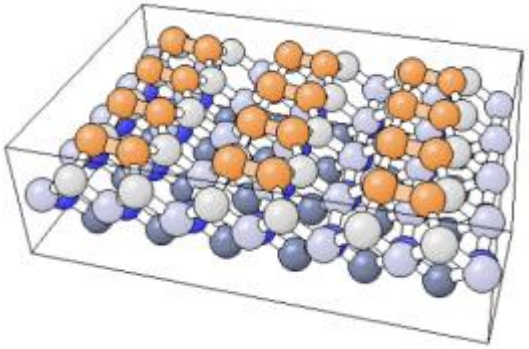
Superlattice



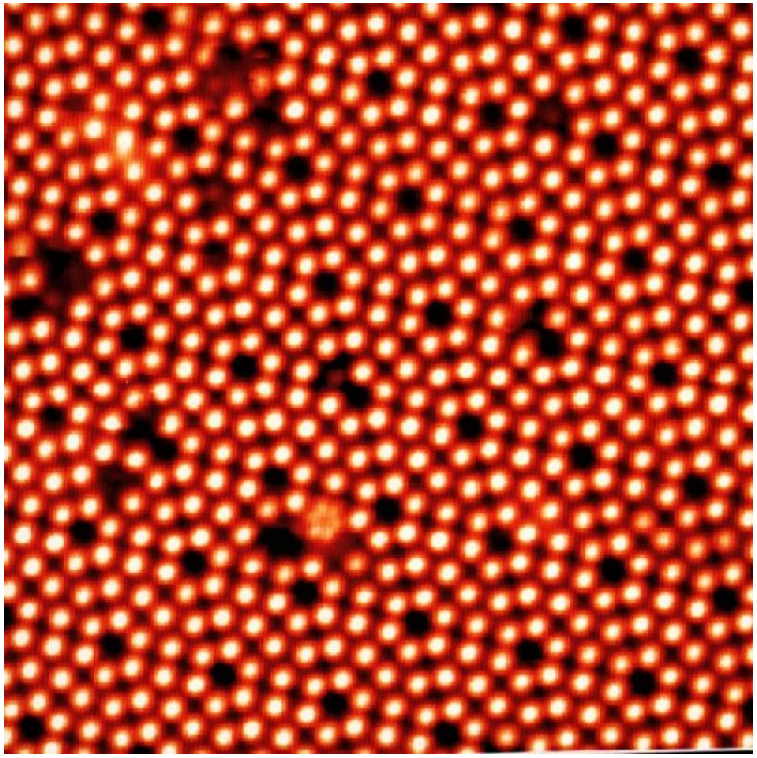
(Image: https://chem.libretexts.org/@go/page/25360)

Examples

Si(100)-(2×1)



Si(111)-(7×7)

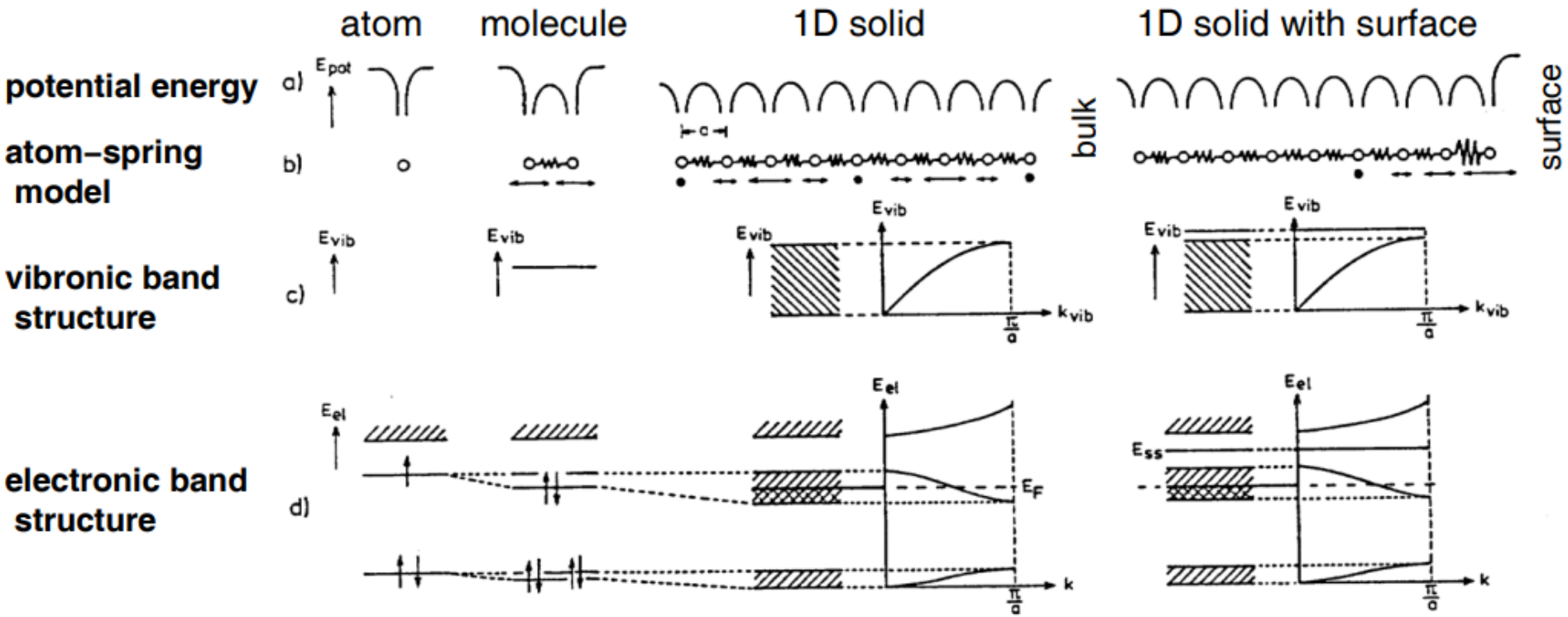


Au(111): herringbone reconstruction



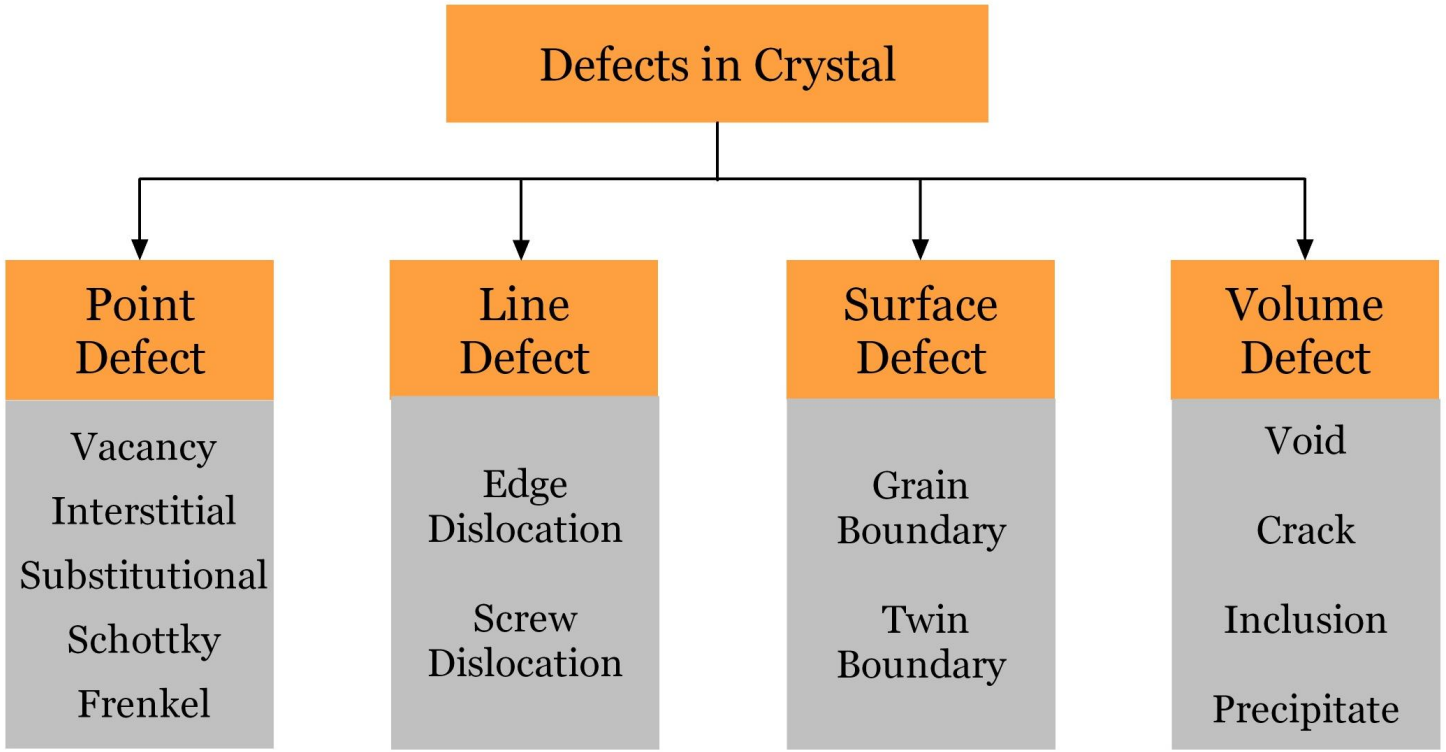
(Images: https://www.chemie-biologie.uni-siegen.de/makro/master_surf_chem/lecture_introsurfchem_1a.pdf
https://www.physik.uni-wuerzburg.de/ep2/media/pre-2014/)

Surface Effects on Band Structures



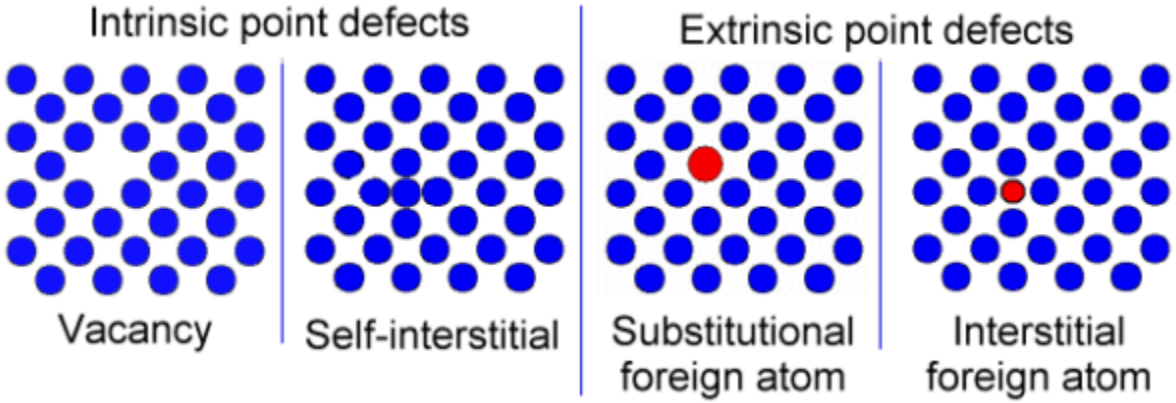
(Image: https://www.chemie-biologie.uni-siegen.de/makro/master_surf_chem/lecture_introsurfchem_1a.pdf)

Defects



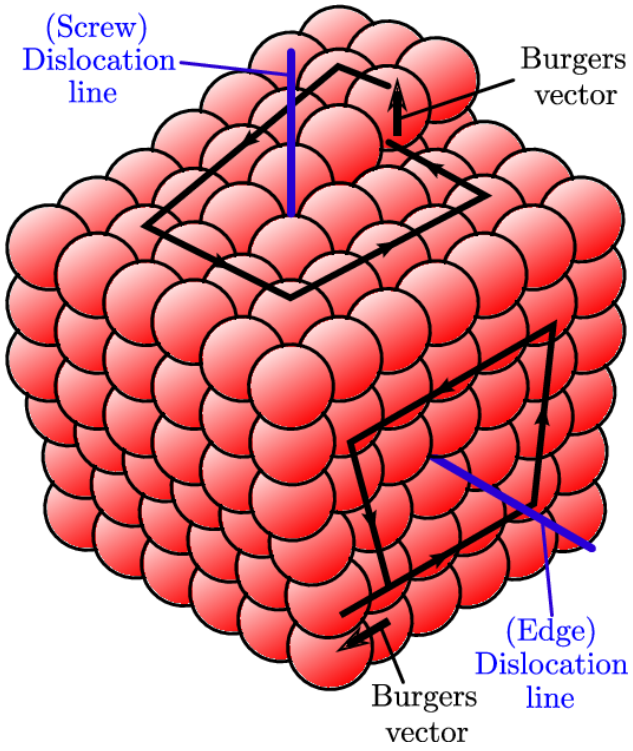
(Image: http://www.minaprem.com/materials-science/defects/vacancy-point-defect-defects-in-solid/)

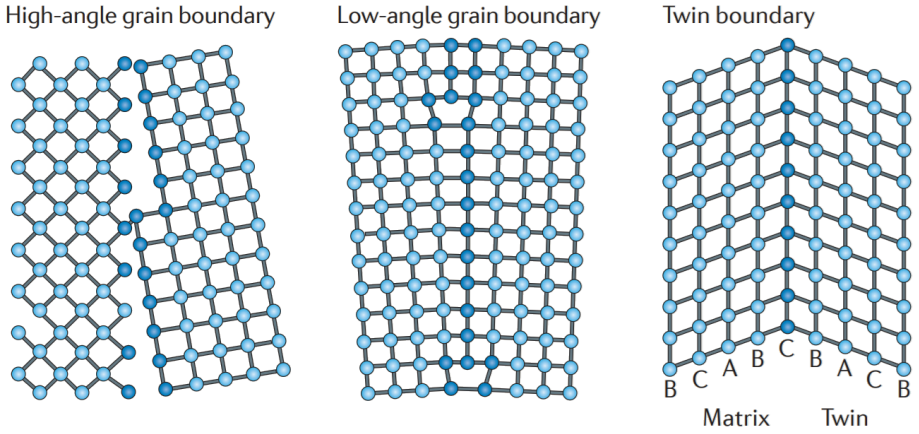
Point Defects



(Image: https://semesters.in/one-two-dimensional-imperfections-in-solids/0-point-defect/)

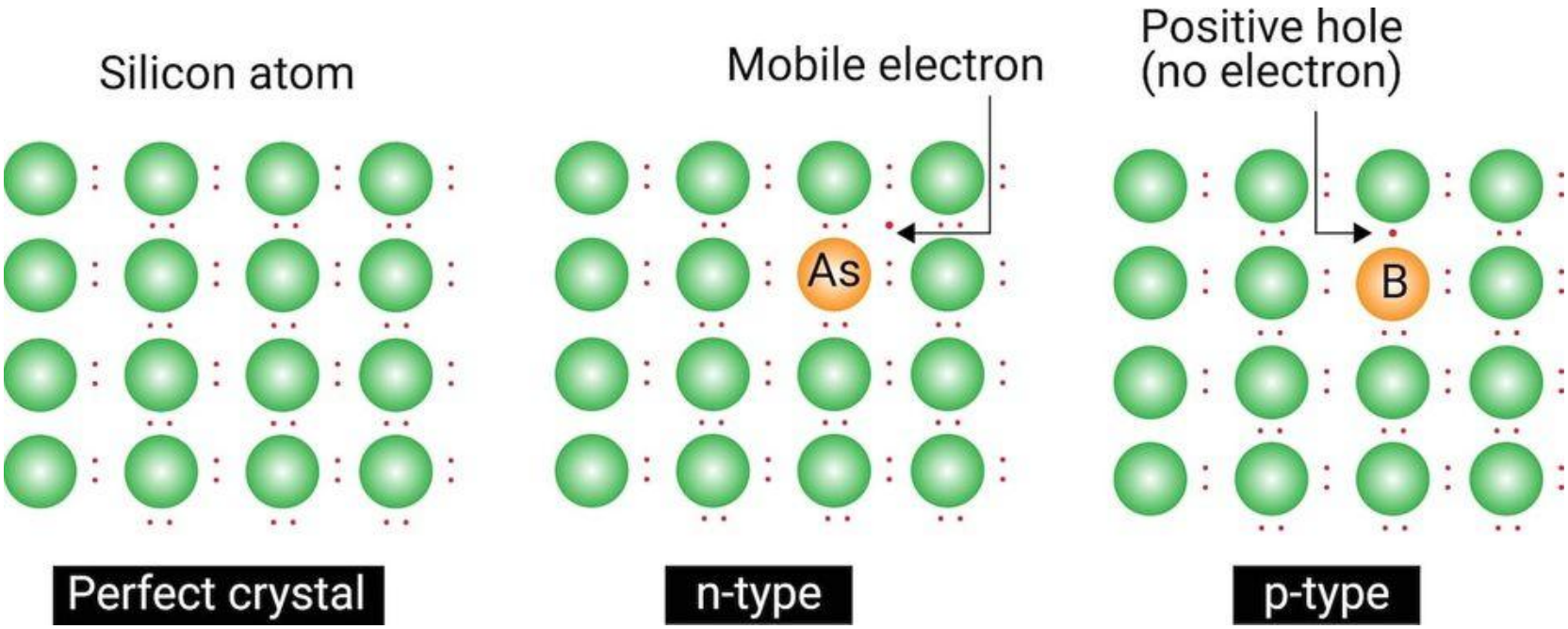
Dislocations & Grain Boundaries





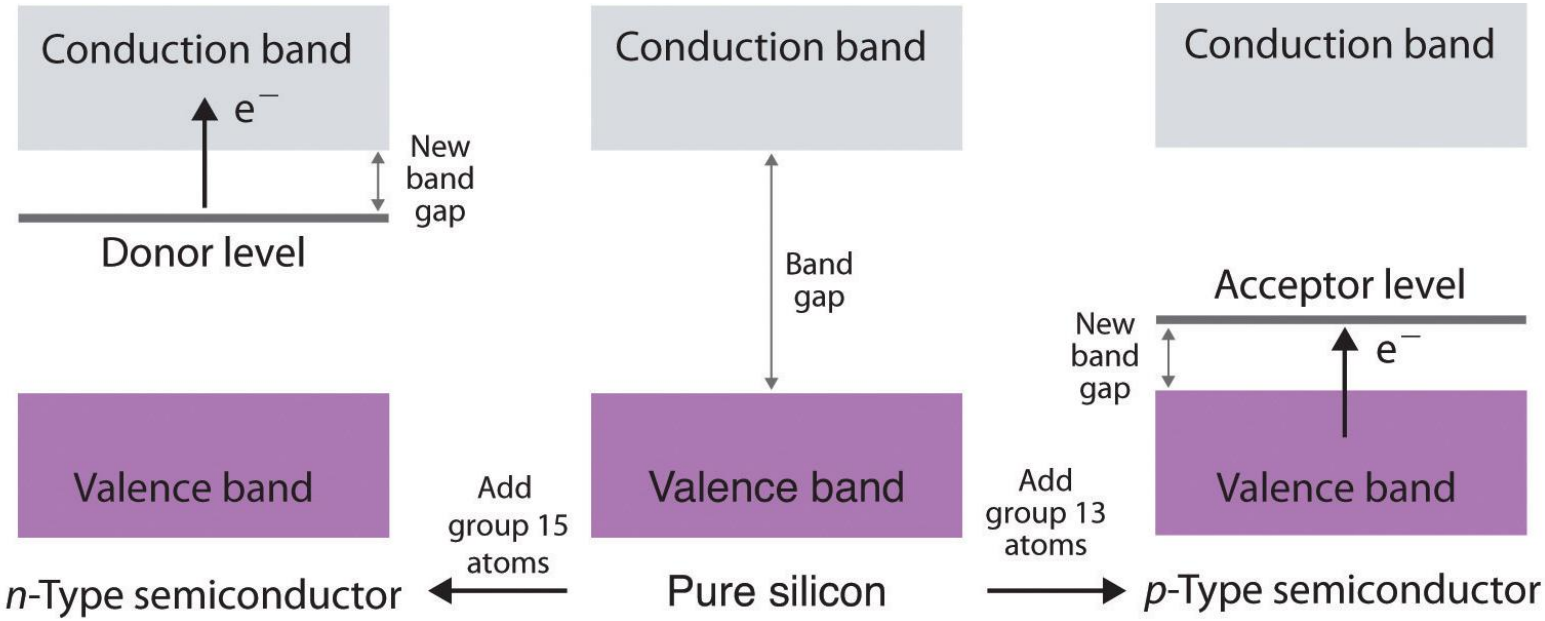
(Image: Hudson, PhD Thesis (2014); Lu, *Nat Rev Mater* (2016))

Defects and Semiconductors



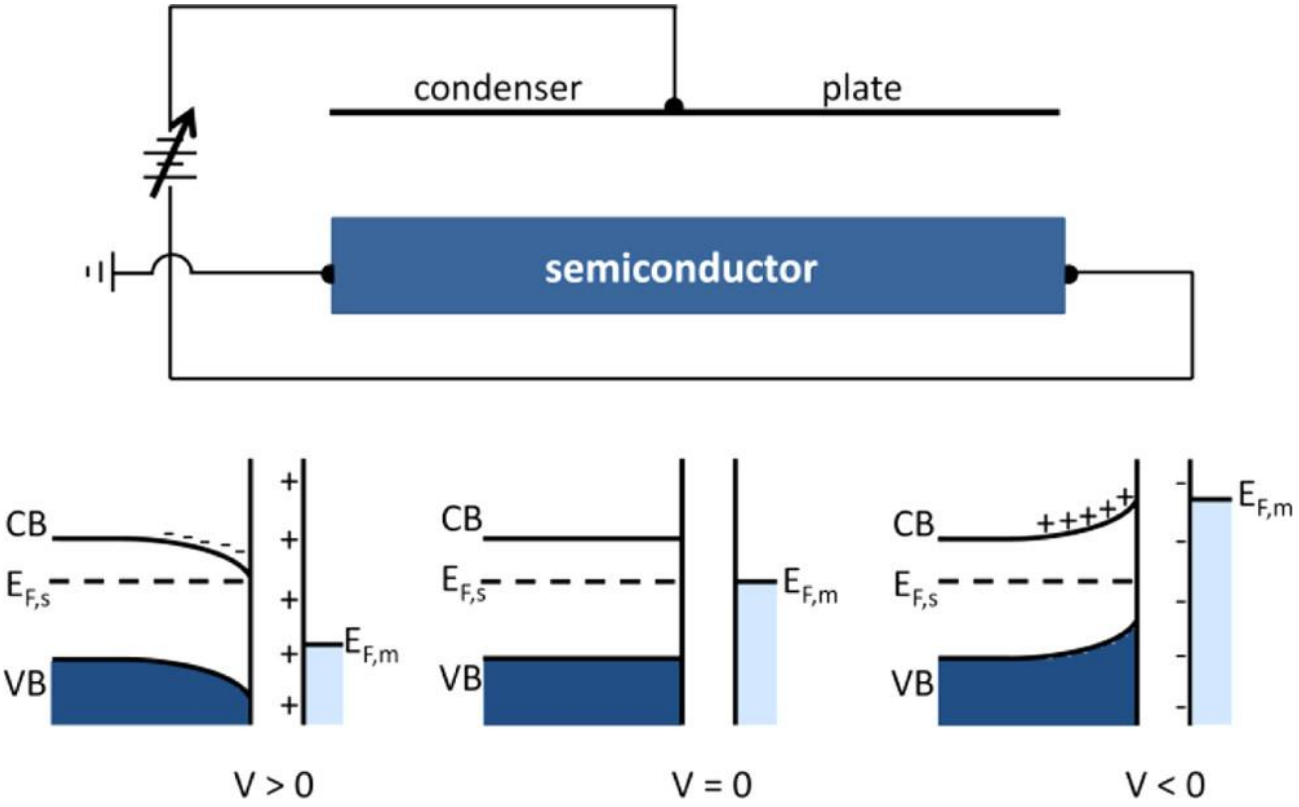
(Image: https://www.wevolver.com/article/understanding-n-type-vs-p-type-semiconductors)

Defects and Semiconductors



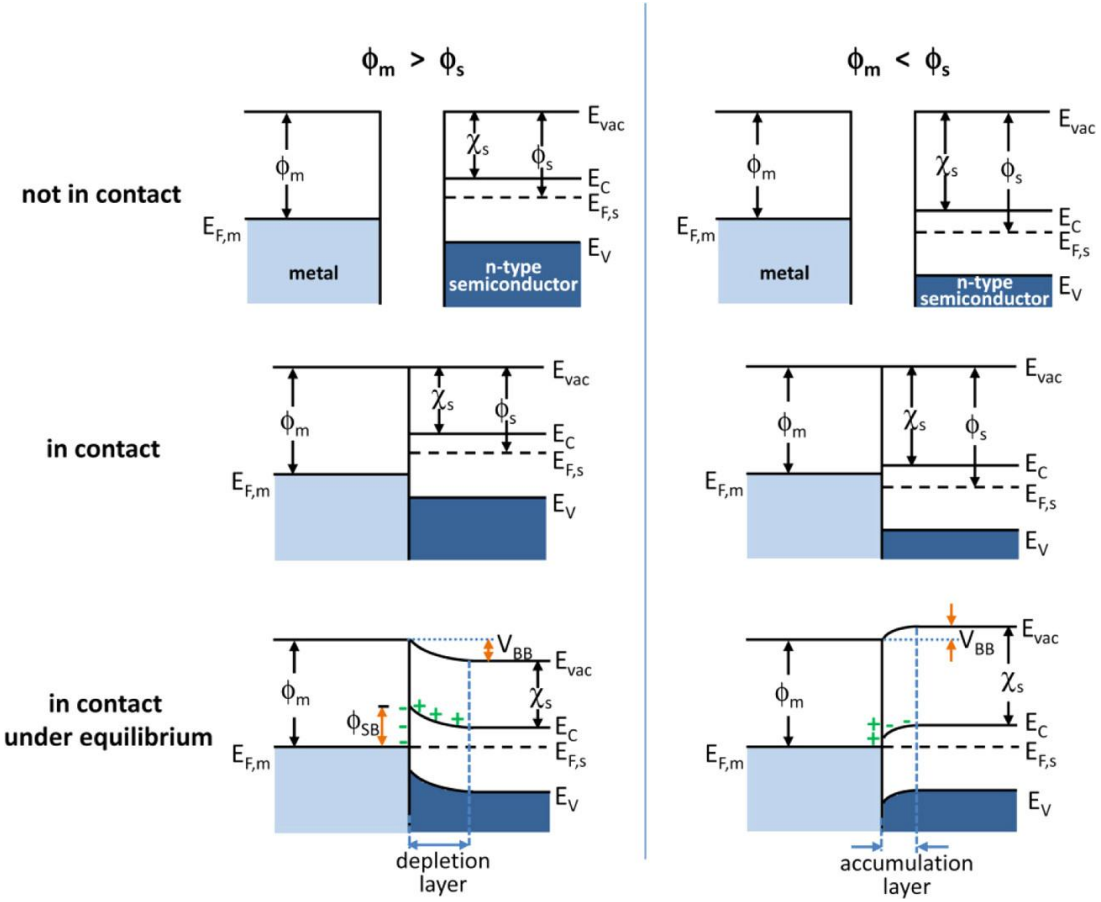
(Image: https://chem.libretexts.org/@go/page/23910)

Band Bending



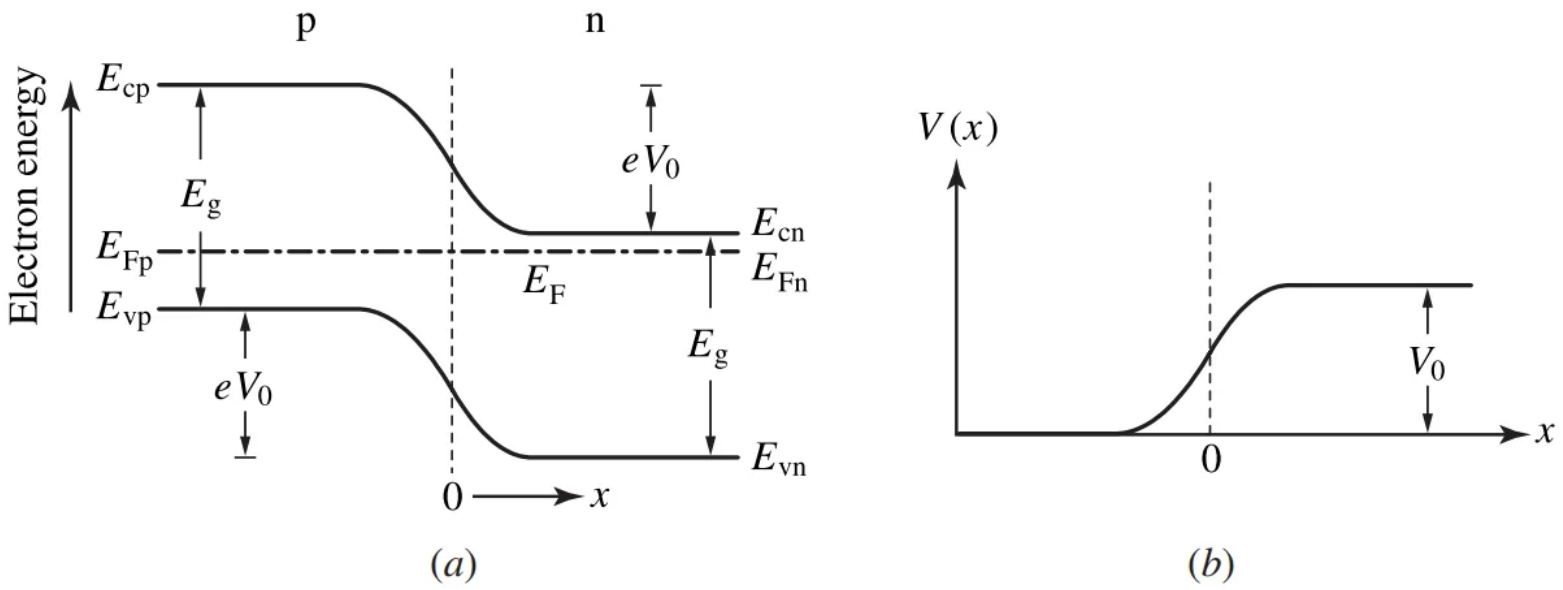
(Image: Zhang and Yates, *Chem Rev* (2012))

Band Bending at Interface



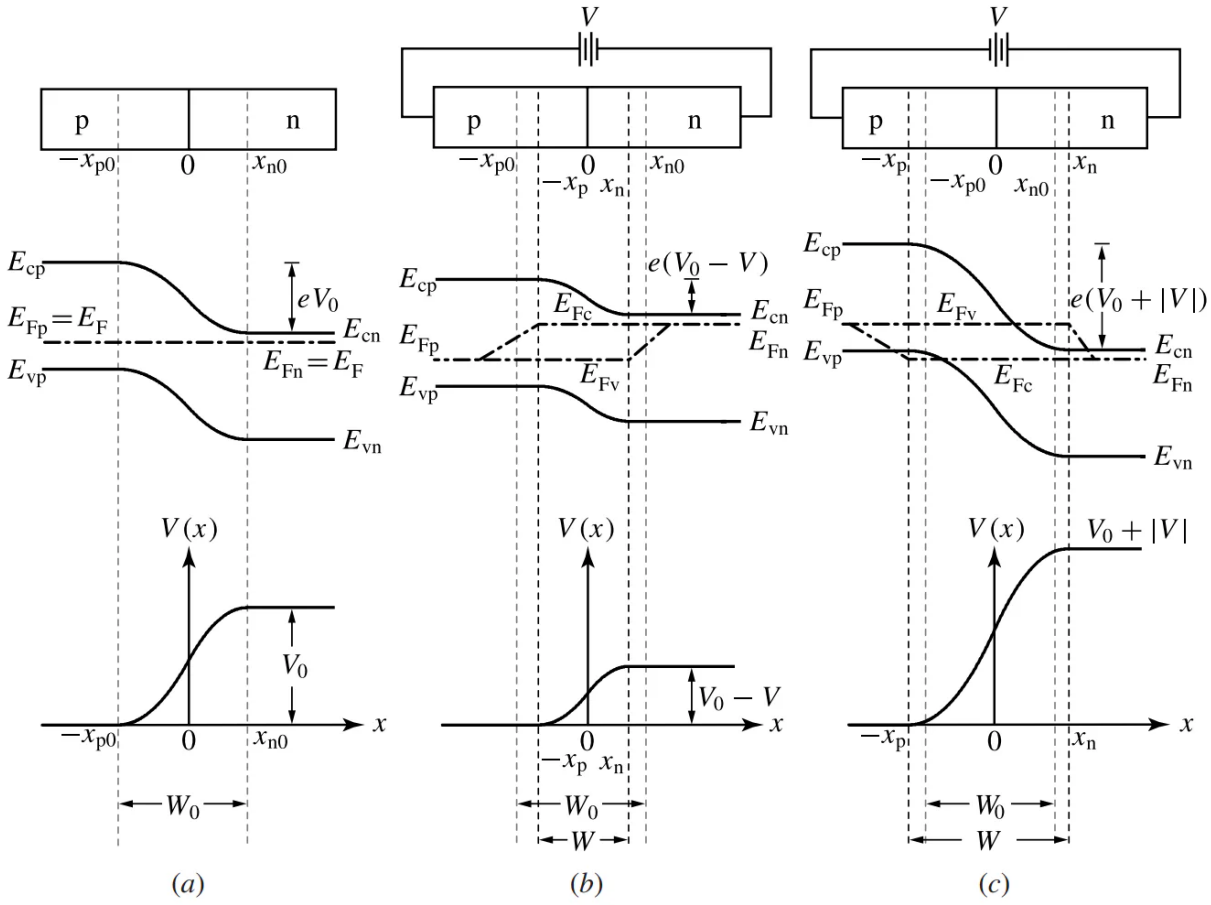
(Image: Zhang and Yates, *Chem Rev* (2012))

p-n Junction



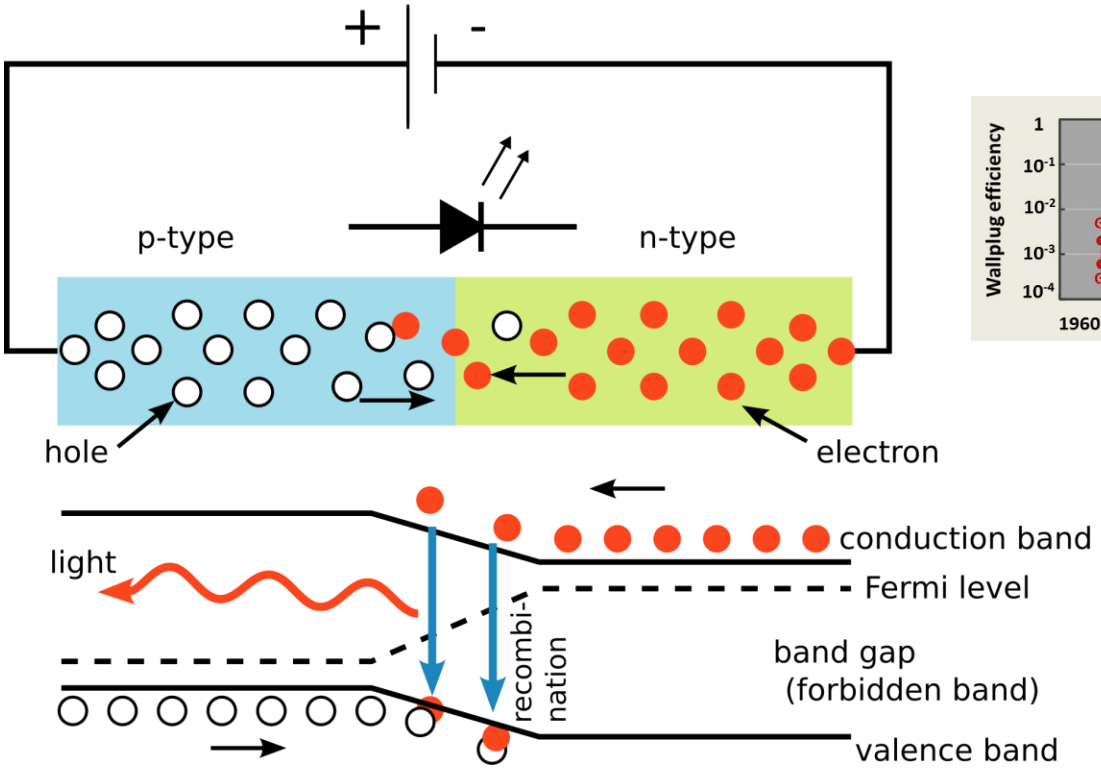
(Image: https://www.fiberoptics4sale.com/blogs/wave-optics/semiconductor-junctions)

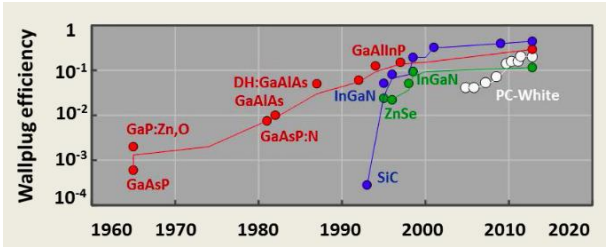
p-n Junction as Rectifier



(Image: https://www.fiberoptics4sale.com/blogs/wave-optics/semiconductor-junctions)

Light-Emitting Diodes (LEDs)



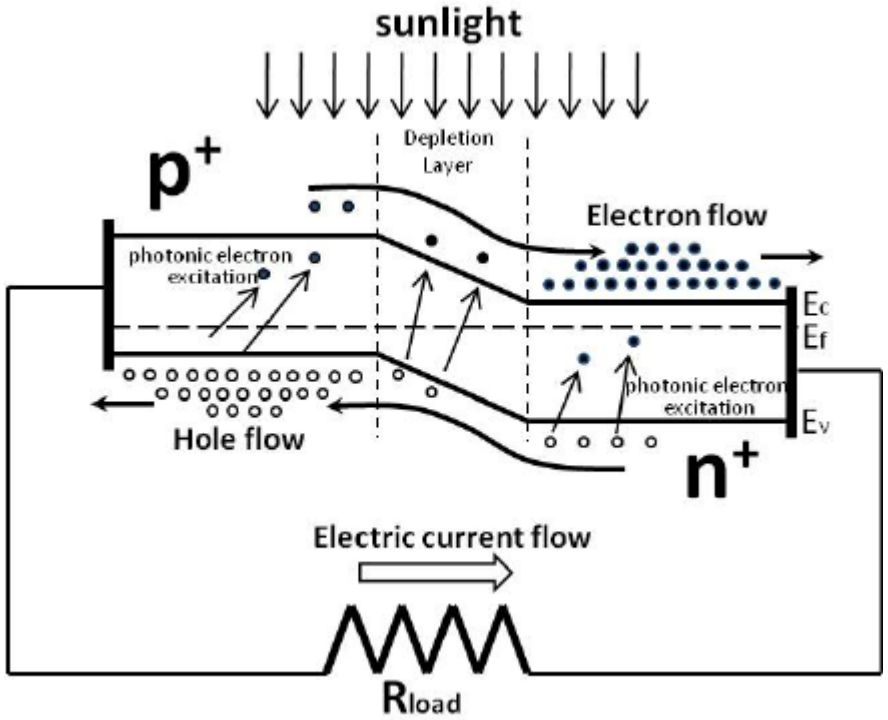




Nobel prize in Physics 2014

(Image: https://commons.wikimedia.org/wiki/File:PnJunction-LED-E.svg)

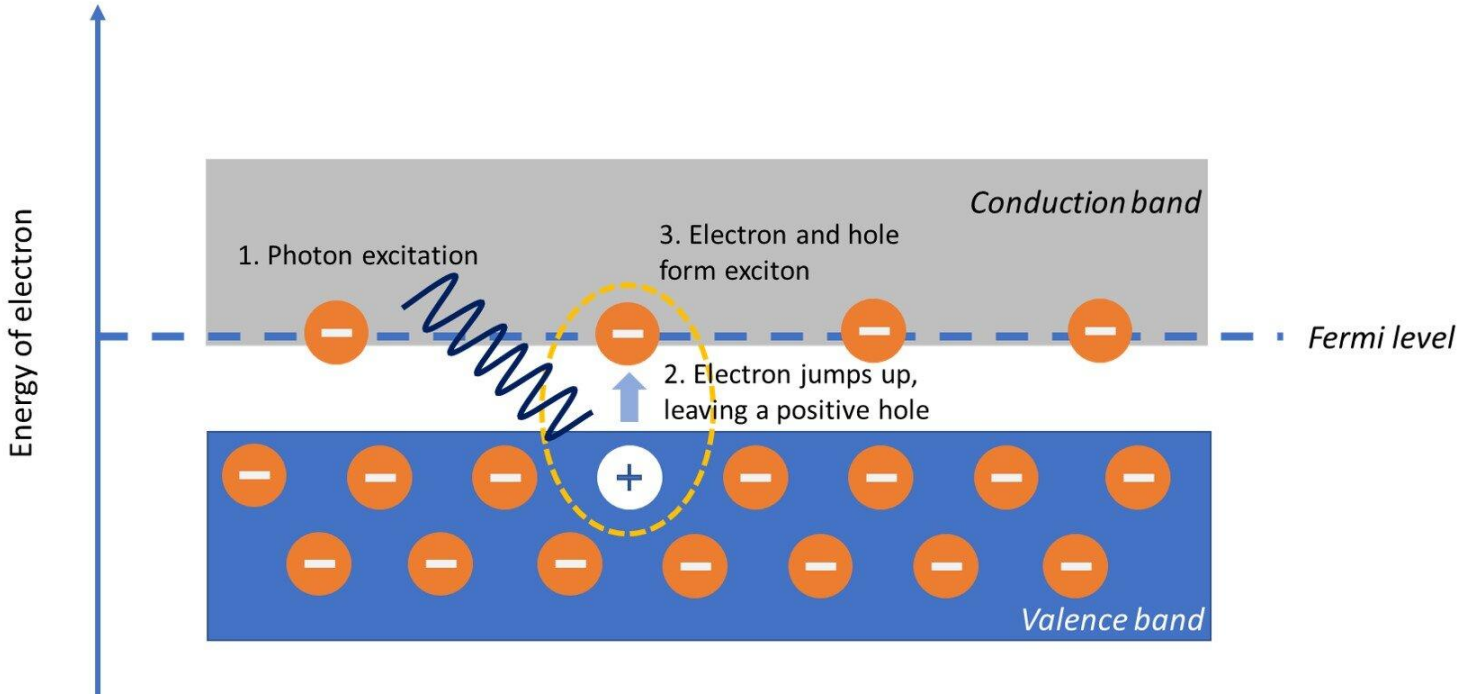
Solar Cells



(Image: https://dx.doi.org/10.5772/19935)

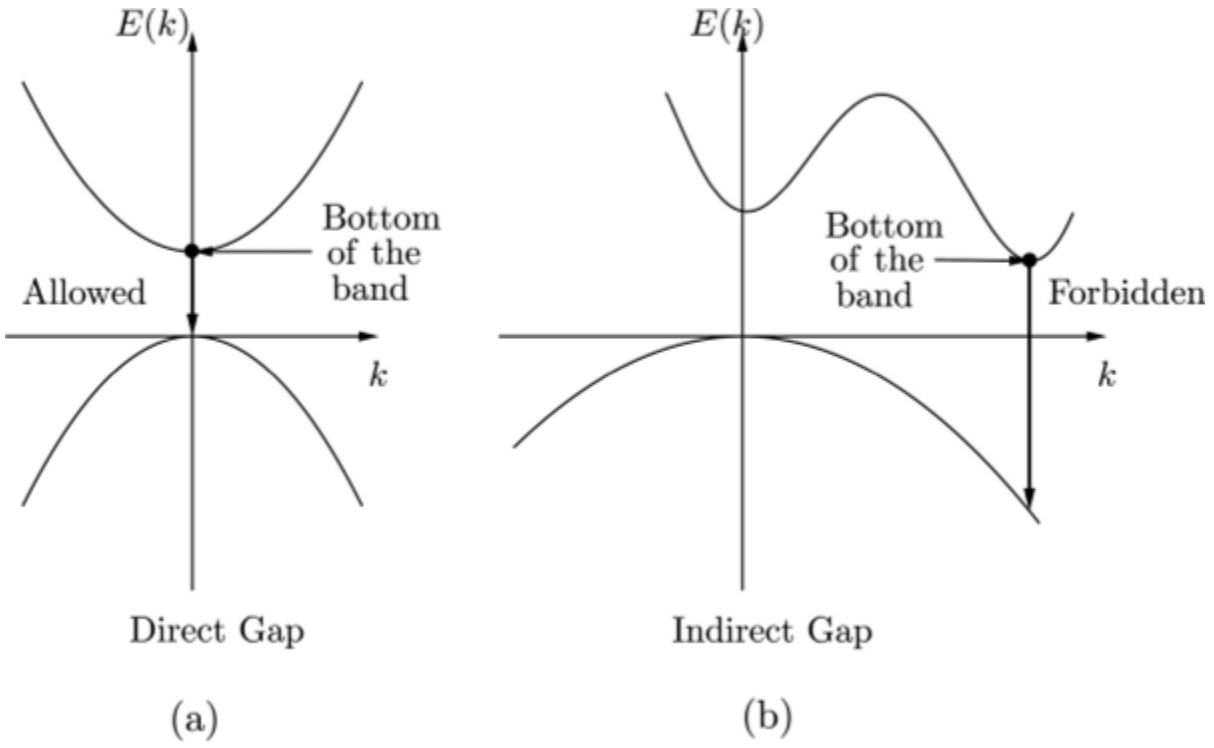
Exciton

(electron-hole coupling)



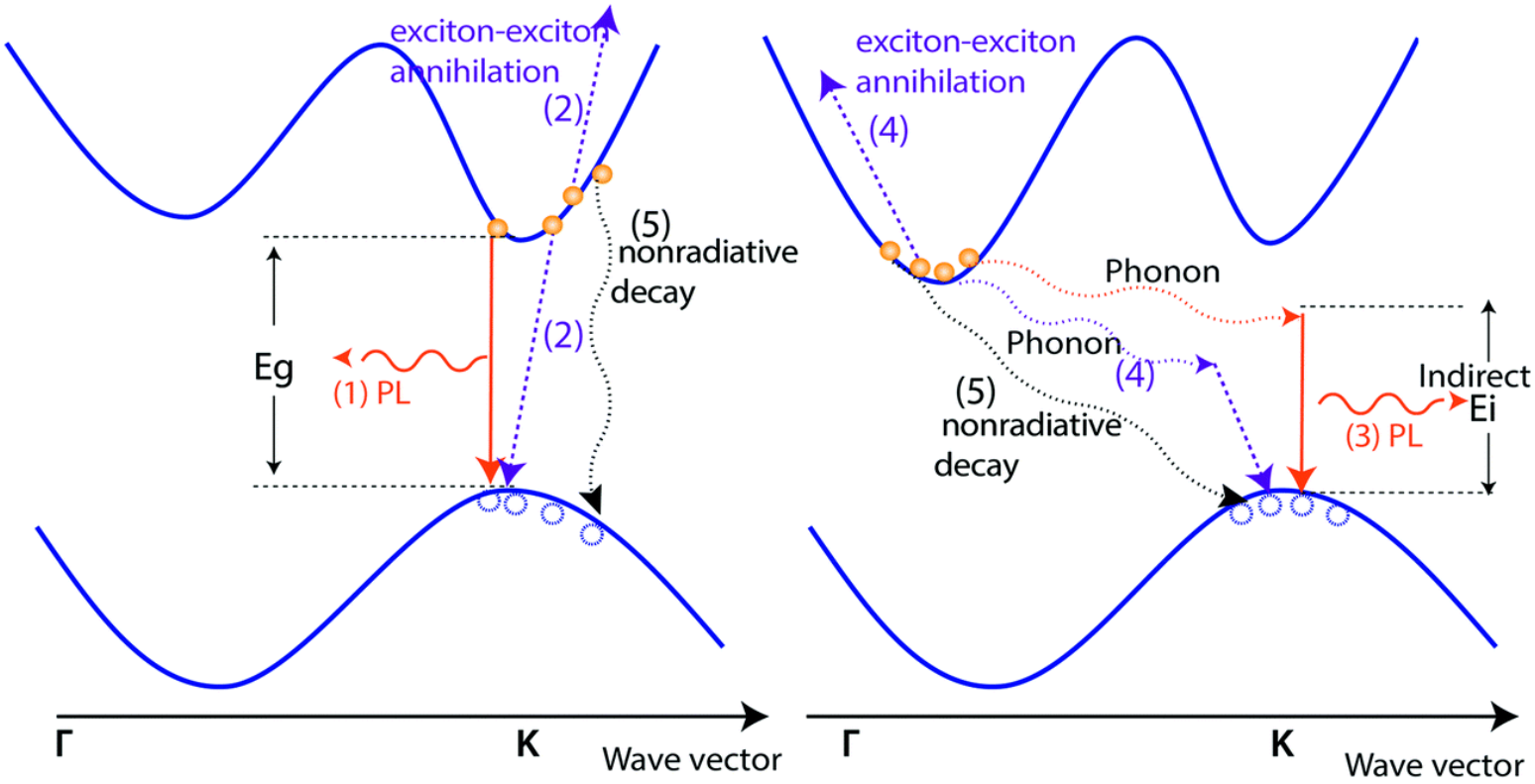
(Image: https://phys.org/news/2022-03-fast-moving-excitons-metal-potential-digital.html)

Direct and Indirect Band Gaps



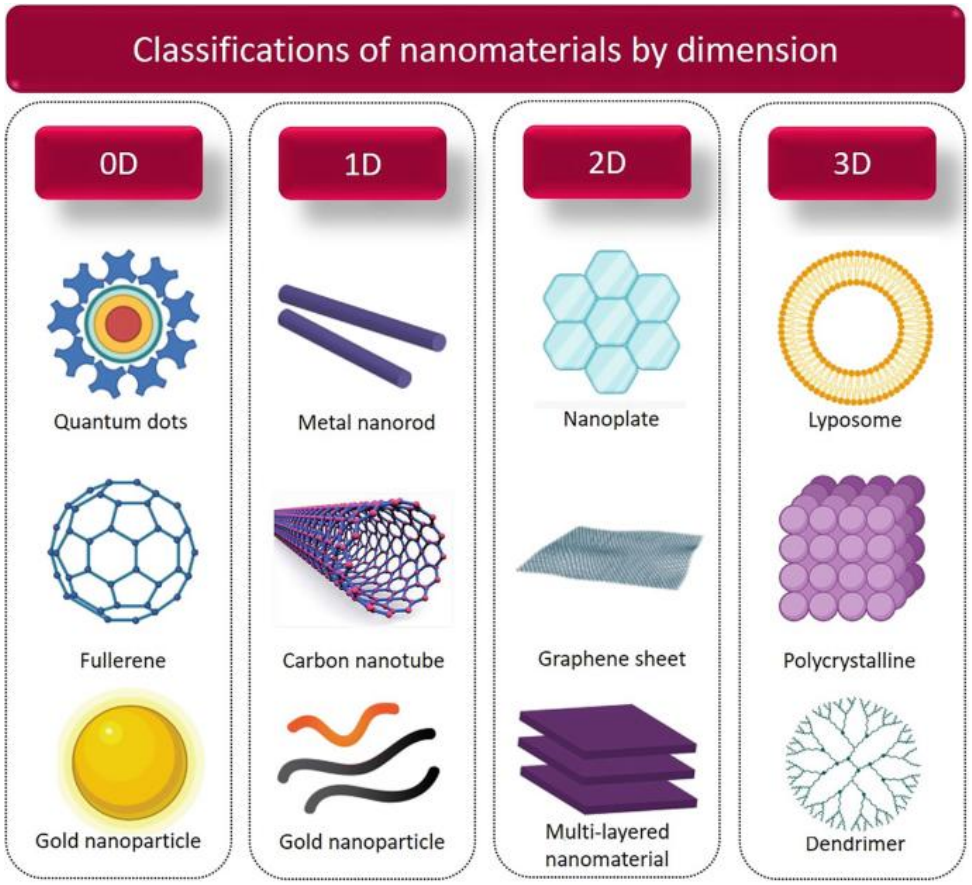
(Image: https://ebrary.net/158530/mathematics/direct_indirect_band_optical_transitions)

Exciton Dynamics



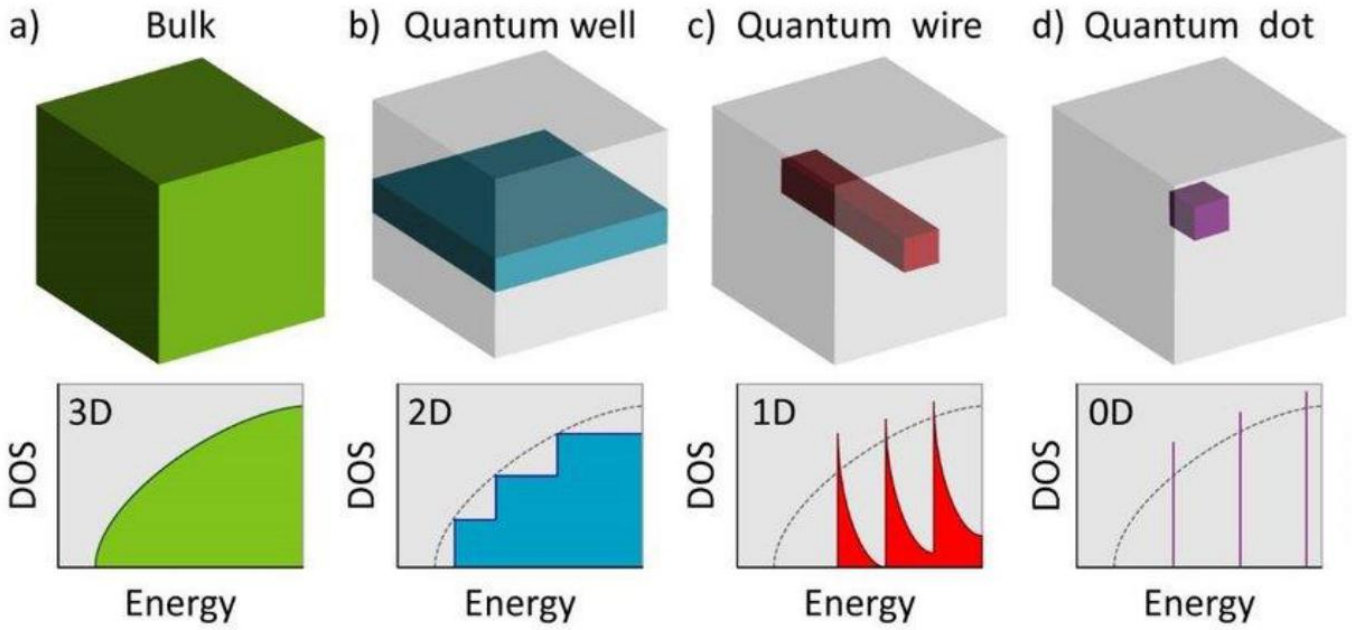
(Image: Yuan and Huang, *Nanoscale* (2015), Fig. 4)

Nanomaterials



(Image: Zainon *et al.*, ***Toxicity of Nanomaterials*** (2022))

Quantum Confinement on DOS



(Image: https://www.nanoinnovation2020.eu/home/PPT/Synthesis%20characteristics%20uses%20and%20detection%20of%20engineered%20nanostructured%20materials.pdf)

DOS of Free Particles

Consider a classical particle in d dimensions. The number of states is

$$N_{\text{states}} = \frac{1}{h^d} \int \mathrm{d}^d \vec{r} \mathrm{d}^d \vec{p} = \frac{V}{h^d} \int \mathrm{d}^d \vec{p}$$

DOS of 3D Free Particles

$$N_{\text{states}} = \frac{1}{h^d}\int \mathrm{d}^d\vec{r}\,\mathrm{d}^d\vec{p} = \frac{V}{h^d}\int \mathrm{d}^d\vec{p}$$

For $d = 3$,

$$\int \mathrm{d}^3\vec{p} = \iiint p^2 \sin\theta \,\mathrm{d}p\mathrm{d}\theta\mathrm{d}\phi = \int 4\pi p^2 \,\mathrm{d}p$$

As $E = p^2/2m$, $p^2\mathrm{d}p = m\sqrt{2mE}\,\mathrm{d}E$ and

$$N_{\text{states}} = \frac{V}{h^3}\int 4\pi m\sqrt{2mE}\,\mathrm{d}E$$

DOS of 3D Free Particles

$$N_{\text{states}} = \frac{V}{h^3} \int 4\pi m \sqrt{2mE}\, \mathrm{d}E$$

On the other hand,

$$N_{\text{states}} = \int g(E)\, \mathrm{d}E$$

Hence,

$$g(E) = \frac{2\pi V}{h^3} (2m)^{3/2} E^{1/2}$$

DOS of 2D Free Particles

$$N_{\text{states}} = \frac{1}{h^d} \int \mathrm{d}^d \vec{r} \mathrm{d}^d \vec{p} = \frac{V}{h^d} \int \mathrm{d}^d \vec{p}$$

For $d = 2$,

$$\int \mathrm{d}^2 \vec{p} = \iint p \, \mathrm{d}p \mathrm{d}\theta = \int 2\pi p \, \mathrm{d}p$$

As $E = p^2/2m$, $p \, \mathrm{d}p = m \, \mathrm{d}E$ and

$$N_{\text{states}} = \frac{A}{h^2} \int 2\pi m \, \mathrm{d}E$$

DOS of 2D Free Particles

$$N_{\text{states}} = \frac{A}{h^2} \int 2\pi m \, \mathrm{d}E$$

On the other hand,

$$N_{\text{states}} = \int g(E) \, \mathrm{d}E$$

Hence,

$$g(E) = \frac{2\pi A}{h^2} m \text{ (constant)}$$

DOS of 1D Free Particles

$$N_{\text{states}} = \frac{1}{h^d}\int \mathrm{d}^d\vec{r}\,\mathrm{d}^d\vec{p} = \frac{V}{h^d}\int \mathrm{d}^d\vec{p}$$

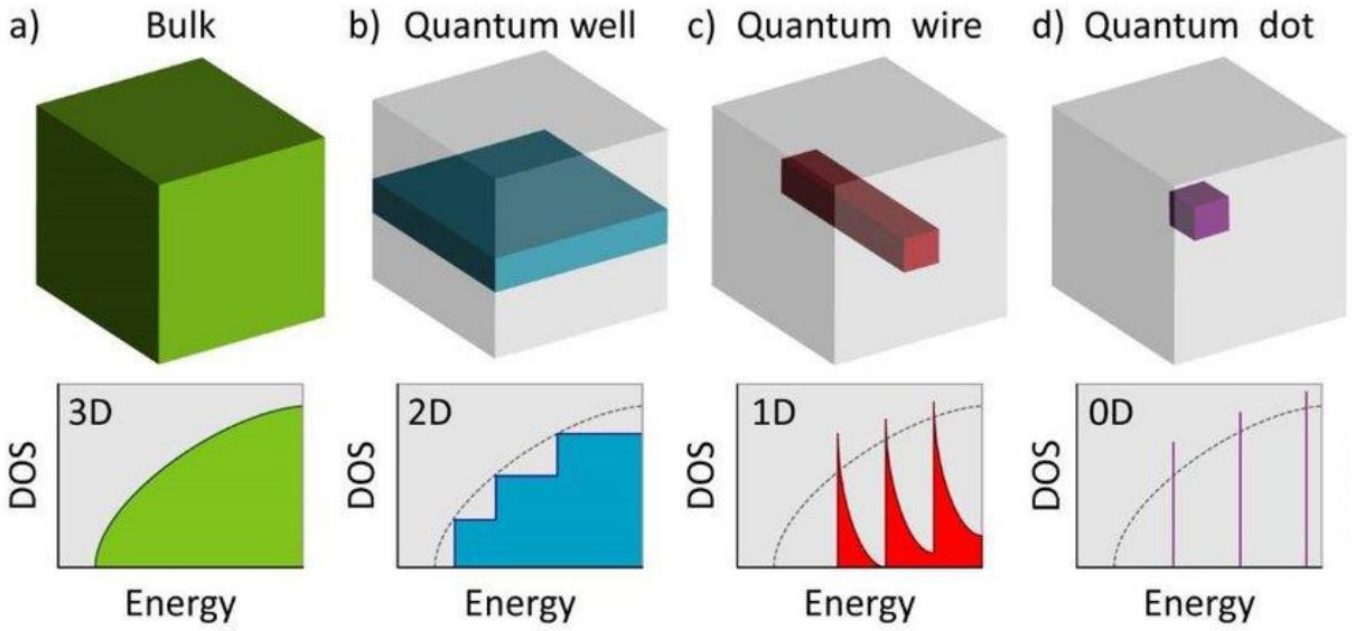
For $d = 1$, as $E = p^2/2m$,

$$\mathrm{d}p = \frac{m}{\sqrt{2mE}}\mathrm{d}E$$

$$N_{\text{states}} = \frac{L}{h}\int \frac{m}{\sqrt{2mE}}\mathrm{d}E$$

$$g(E) = \frac{Lm}{h\sqrt{2mE}}$$

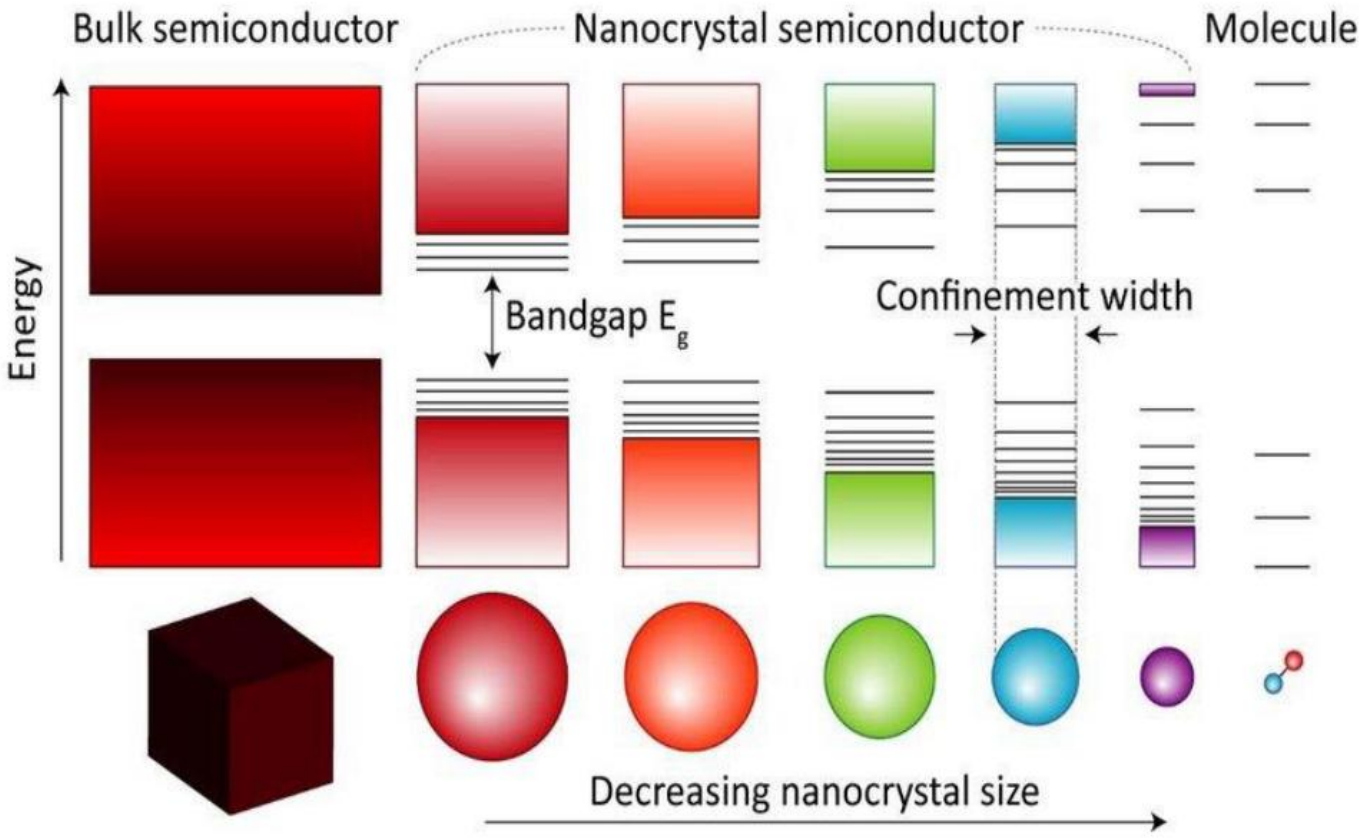
Quantum Confinement on DOS



$g(E) \propto E^{1/2}$ $g(E) \propto E^{0}$ $g(E) \propto E^{-1/2}$

(Image: https://www.nanoinnovation2020.eu/home/PPT/Synthesis%20characteristics%20uses%20and%20detection%20of%20engineered%20nanostructured%20materials.pdf)

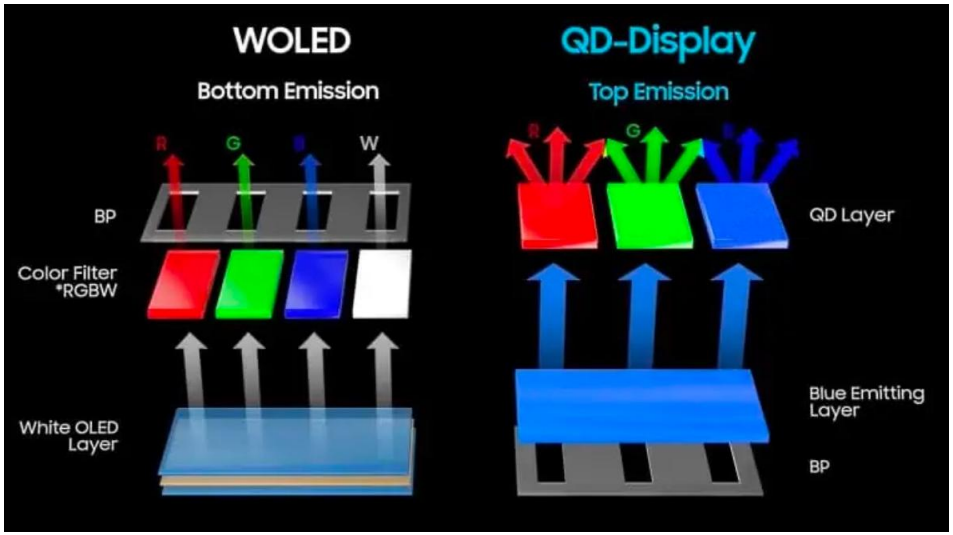
Quantum Dots: Size Matters



(Image: https://www.nanoinnovation2020.eu/home/PPT/Synthesis%20characteristics%20uses%20and%20detection%20of%20engineered%20nanostructured%20materials.pdf)

Application of Quantum Dots







Nobel prize in Chemistry 2023

(Images: https://avantama.com/what-is-a-quantum-dot/
https://www.sammobile.com/news/qd-oled-tv-features-explained/)